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LESSON PLAN	**COURSE: MHF4U**
Date: Lesson #5	**Unit of Study: Chapter #3 Rational Functions**
Materials / Resources: - **Advanced Function 12 Study Guide** - **Chapter #3 Pre-assessment Quiz** - **Chapter #3 Formula, Aid and Homework Sheet(s)**	**Lesson Focus** - **Reciprocal of a linear function** - **Reciprocal of a quadratic function** - **Rational functions of the form** $f(x) = \frac{ax+b}{cx+d}$

Prior Knowledge:

- Rational functions in the form of $f(x) = \frac{1}{x}$
- Definitions of asymptotes and their relationship to undefined denominators

Overall Expectations:

1. Identify and describe some key features of the graphs of rational functions, and represent rational functions graphically
2. Solve problems involving polynomial and simple rational equations graphically and algebraically
3. Demonstrate an understanding of solving polynomial and simple rational inequalities

Specific Expectations:

2.1 Determine, through investigation with and without technology, key features (i.e., vertical and horizontal asymptotes, domain and range, intercepts, positive/negative intervals, increasing/decreasing intervals) of the graphs of rational functions that are the reciprocals of linear and quadratic functions, and make connections between the algebraic and graphical representations of these rational functions [e.g., make connections between $f(x) = \frac{1}{x^2-4}$ and its graph by using graphing technology and by reasoning that there are vertical asymptotes at $x = 2$ and $x =- 2$ and a horizontal asymptote at $y = 0$ and that the function maintains the same sign as $f(x) = x^2 - 4$]

2.2 Determine, through investigation with and without technology, key features (i.e., vertical and horizontal asymptotes, domain and range, intercepts, positive/negative intervals, increasing/decreasing intervals) of the graphs of rational functions that have linear expressions in the numerator and denominator [e.g., $f(x) = \frac{2x}{x-3}$, $h(x) = \frac{x-2}{3x+4}$], and make connections between the algebraic and graphical representations of these rational functions

2.3 Sketch the graph of a simple rational function using its key features, given the algebraic representation of the function

3.5 Determine, through investigation using technology (e.g., graphing calculator, computer algebra systems), the connection between the real roots of a rational equation and the x-intercepts of the graph of the corresponding rational function, and describe this connection [e.g., the real root of the equation

$\frac{x-2}{x-3} = 0$ is 2, which is the x-intercept of the function $f(x) = \frac{x-2}{x-3}$; the equation $f(x) = \frac{1}{x-3}$ has no real roots, and the function $f(x) = \frac{1}{x-3}$ does not intersect the x-axis]

3.6 Solve simple rational equations in one variable algebraically, and verify solutions using technology (e.g., using computer algebra systems to determine the roots; using graphing technology to determine the x-intercepts of the graph of the corresponding rational function)

3.7 solve problems involving applications of polynomial and simple rational functions and equations [e.g., problems involving the factor theorem or remainder theorem, such as determining the values of k for which the function $f(x) = x^3 + 6x^2 + kx - 4$ gives the same remainder when divided by $x - 1$ and $x + 2$]

Learning Goals:

- Students will learn how to get the equations of the vertical and horizontal asymptotes from given rational functions
- Students will know how to determine the x- and y-intercept of given rational functions
- Students will be able to conclude the domain and range of given rational functions
- Students will learn how to sketch and label rational functions

Success Criteria:

1. By the end of this lesson I will be able to determine the equations of the vertical and horizontal asymptotes of simple rational functions
2. By the end of this lesson I will be able to determine the x- and y-intercept of simple rational functions
3. By the end of this lesson I will be able to determine the domain and range of simple rational functions
4. By the end of this lesson I will be able to sketch and label graphs of simple rational functions

Assessment Strategies:

<u>Diagnostic</u>: chapter #3 pre-assessment quiz

<u>Formative</u>: homework for chapters 3.1, 3.2 & 3.3

<u>Summative</u>: quiz #3, test #2

Time: **Part 1**	**Introduction:** Students will be given a pre-assessment quiz to determine their knowledge standings for rational functions, and then proceed from their answers.
Time: **Part 2**	**Lesson:** <u>Chapter 3.1 Reciprocal of a Linear Function</u> Determine the form of a reciprocal of a linear function. Establish the characteristics of a reciprocal of a linear function such as its x- and y-intercept and its vertical and horizontal asymptotes. Conclude since the numerator of this function will always be a constant (hence *reciprocal* of a linear function) that it will never have an x-intercept

<table>
<tr><td></td><td>because the numerator will never equal to zero, and thus $f(x)$ will never equal to zero. Also determine that the vertical asymptote will be the root of the denominator and the horizontal asymptote for a reciprocal of a linear function will always be $y = 0$. Give examples by graphing certain functions.

<u>Chapter 3.2 Reciprocal of a Quadratic Function</u>
Determine the form of a reciprocal of a quadratic function. Establish the characteristics of a reciprocal of a quadratic function such as its x- and y-intercepts and its vertical and horizontal asymptote. Conclude that like a reciprocal of a linear function, since the numerator of this function will also always be a constant that it will never have an x-intercept because the numerator will never equal to zero and thus $f(x)$ will never equal to zero. Also determine that the vertical asymptote will be the roots of the denominator which in this case, there could be 2 and the horizontal asymptote for a reciprocal of a quadratic function will also always be $y = 0$. Give examples by graphing certain functions.

<u>Chapter 3.3 Rational Functions of the Form</u> $f(x) = \frac{ax+b}{cx+d}$
With the rational function of the form $f(x) = \frac{ax+b}{cx+d}$, again establish the characteristics of this form of a rational function such as its x- and y-intercepts and its vertical and horizontal asymptotes. For this form of rational functions, conclude that there are x-intercepts when a satisfied x makes the numerator equal to 0. Also determine that the vertical asymptote is when $x =- \frac{d}{c}$ and the horizontal asymptote equations to the coefficient of the leading variable in the numerator divided by the coefficient of the leading variable in the denominator, e.g. $y = \frac{a}{c}$. Give examples by graphing certain functions.</td></tr>
<tr><td>Time:

Part 3</td><td>Wrap-up:

Ask students to demonstrate their acquired knowledge by applying them to example problems and then answer any unresolved questions.</td></tr>
<tr><td colspan="2">Notes, reminders & homework:

<u>Reminder</u>:
- Quiz #3 (to be written before / after lesson #6)

<u>Homework</u>:
Chapter 3.1: #1 – 4, 6 – 9, 20
Chapter 3.2: # 1 – 11
Chapter 3.3: # 1 – 9, 13, 14 – 16</td></tr>
<tr><td colspan="2">Lesson Reflection:</td></tr>
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